

Mechanics Before Preference

A Faithful, Formally Verified Quality Standard for the Souls of Onmyoji

KCN-JUDU

Yata Quality Model v2 (yata-quality-v2); formal development KCN-judu/yata, formal/ (Lean 4.34.0, Mathlib v4.34.0); paper revision of 2026-09-25

ABSTRACT

A soul (御魂) in *Onmyoji* is produced by a random process that the game’s numerical designers specified and, in large part, published: which sub-attributes a soul starts with, which attribute each enhancement strengthens, and within which range each increment falls. A quality score for souls is only as trustworthy as its relation to that process. This paper presents the Yata Quality Model, a scoring standard organized around one principle, **mechanics before preference**: the standard first reconstructs the designers’ generative process as a probability measure, taking every probability from a published rule and fitting nothing to data, and only then applies a small number of declared adjustments that make the numbers read the way players read souls. The reconstruction is exact: the law of how many increments reach an attribute set is computed by rational propagation, and the one quantity the designers did not publish — the distribution of an increment within its range — enters every score only through its mean, which three independent sources constrain. The adjustments measure value in *roll units*; judge a soul against a published catalogue of role *archetypes* — output, hit, healing, and a one-attribute speed archetype that is considered only once a soul’s Speed reaches four roll units; anchor each archetype’s score at zero, at its own expected soul and at its own attainable maximum; and cut tiers N to UR at stated utility milestones. Archetypes need not have the same size: comparability comes from each archetype’s own normalization. We prove in Lean 4 that none of the adjustments feeds rarity back into the score, that the score is bounded, monotone and dominance-preserving, that the healing archetype is the output archetype under a relabelling the published rules respect, that tier assignment is deterministic with the stated precedence, that one perfect line alone is exactly SSR, that every UR soul outscores every other soul of its archetype, and that the speed archetype’s gate and tiers mean what they say. A calibration program recomputes every number in this paper and fails the build when a published number drifts.

Keywords: game mechanics, probabilistic modelling, scoring standards, normalization, exact computation, mechanized proof, Lean 4

1 INTRODUCTION

A player looking at a soul asks two questions: *is this soul good?* and *is it good for this Shikigami?* Yata answers them separately. This paper is about the first, the soul’s quality with no holder in mind, and about a property that a quality score has to earn before any of its numbers deserve trust: that it measures the soul against the game the designers made, not against a model the scorer invented.

That property is not automatic. A score is a function of a soul, and a normalization is a function of a reference distribution; if the reference distribution is fitted to a sample, or assumed for convenience, or tuned until the tiers feel right, then the score reflects those choices as much as the game. The soul system invites this, because it has parts the designers published and a part they did not. The published parts fix which attributes a soul can hold and with what probabilities they are drawn and strengthened. The unpublished part is how large an increment is within its published range.

We therefore separate the standard into two layers and keep them apart throughout.

- **(L1) The mechanics layer** is the designers' process as a probability measure. Every probability in it is a published fraction or a published rule; nothing in it is estimated from observed souls; it is propagated exactly. Where the designers published nothing, it depends on as little as possible and says how much.
- **(L2) The reading layer** is the standard's own: how a soul's values become one number and one tier that a player can read. Every choice in it is a declared adjustment with a stated reason, and none of them alters the measure of L1.

The requirements on the standard follow from this split, and we label them so that the rest of the paper can name the requirement each mechanism answers.

- **(R1) Faithful.** Probabilities come from the published rules, and the reference quantities are computed exactly from them.
- **(R2) Robust where the designers were silent.** A score depends on unpublished quantities only through what three independent readings constrain.
- **(R3) No reward for rarity.** A soul is not scored higher because the outcome it shows is improbable.
- **(R4) A common scale.** Souls judged against different roles are placed on one 0–100 scale without favouring the role whose attributes are drawn less often, whatever the number of attributes the role values.
- **(R5) Tiers mean what their words say.** The player-facing name of a tier and its formal predicate coincide, and the tier order cannot be contradicted by the score order where the standard promises it cannot.

1.1 Contributions

1. **The object of analysis** (§2). We fix exactly which part of the game the standard models — the six-star enhancement system — and which sources each rule rests on, under a stated precedence: an official publication first, then the newest analysis, then larger samples.
2. **A faithful reconstruction** (§3, R1–R2). The generative process of the published rules is a finite probability measure over hit vectors, computed exactly in rational arithmetic. Proposition 1 shows that expected utility depends on the unpublished increment law only through its mean; the reconstruction reproduces an exact symmetry check, the analysis's theoretical best soul, and a Monte Carlo simulation within two and a half standard errors.
3. **Declared adjustments for players** (§4–§7, R3–R4). Roll units; a catalogue of archetypes of different sizes; main-attribute gating and a soul-level eligibility predicate; three anchors per archetype; and utility-milestone tiers. Theorem 6 shows that probability enters a score only through the archetype's expected utility; Theorem 7 that archetypes the published rules treat alike share one scale exactly, and that healing and output are such a pair; the residual between the others is measured.
4. **Tier semantics** (§6–§7, R5). Theorems 8–13 prove that the four-attribute ladder is deterministic with precedence UR, SP, band; that SP cannot bypass its floor; that one perfect line alone is exactly SSR; that UR is above every other soul of its archetype in score and in the exposed order; that dominance never lowers a tier; and that the v1.1 revision of the SSR edge only raised tiers. Proposition 14 and Theorems 15–16 do the same for the one-attribute speed archetype: which souls it may judge, its scale, and its tier rule.
5. **Calibration and reproducibility** (§8–§9). A calibration program computes every number here — anchors, thresholds, tier rates under three increment laws and two readings of the weights, the eligibility rate of speed, a Monte Carlo check, and thirty golden test vectors — and a repository check fails when a committed number drifts.

Every theorem is a Lean 4 declaration in `formal/lean/ of KCN-judu/yata` [4, 5], cited by its Lean name where it is stated and indexed in Appendix A; Proposition 1 and the statements marked *computation* are established by the paper's argument and by the calibration program, not in Lean. Two conventions bound every claim. A theorem proves consequences of the stated definitions; that the game follows the stated rules rests on the sources of §2, not on Lean. And the standard makes no claim about what players prefer: its adjustments are designed to match how players read souls, and whether they do is a question no theorem answers.

This paper explains `yata-quality-v2`. Version 1.1 moved the SSR edge (§6.4); version 2 replaced the catalogue (§4.2, §7).

2 THE OBJECT OF ANALYSIS

2.1 What is modelled

The object is the **six-star soul at +15** and the random process that produces its sub-attributes. A soul has a slot (1–6), a main attribute from that slot’s pool, a level, and up to four sub-attributes drawn from eleven: `AtkFlat`, `AtkPercent`, `DefFlat`, `DefPercent`, `HpFlat`, `HpPercent`, `Spd`, `EffectHit`, `EffectRes`, `Crit`, `CritDmg`. A sub-attribute’s value is the sum of its *increments*: the initial one, and one for each enhancement that strengthens it.

The following are outside the object, and the standard says nothing about them: souls below six stars, whose increment ranges are not published; set effects, which belong to the second, need-dependent pass of Yata’s scoring; the innate attribute of boss souls; and how often each slot or main attribute is dropped by a given activity, which the designers disclose only for four shop sources.

2.2 The published rules

The rules are those of Yata’s mechanics specification, each with its source. Table 1 lists the ones the standard uses.

Rule	Source	Status
a soul starts with 2, 3 or 4 sub-attributes, about one third each	community data-mining cited by [2]	secondary; the cited forum requires a login
at +3, +6, +9, +12 and +15, one enhancement	[2]; official guides	primary
with four sub-attributes, each is strengthened with probability 1/4	[6]	official
below four, one attribute is drawn from all eleven; a new one is added, an existing one strengthened	[6]; [2]	official weights; the strengthening reading from the analysis
class weights 攻击类 36%, 防御类 36%, 功能类 28%, shared equally within a class	[6]; the equal share [2]	official
one increment of attribute a lies in $[0.8 h_a, h_a]$, h_a its largest increment	[2], Table 4	primary for six stars

Table 1. The rules of the six-star enhancement process and their sources.

Precedence of sources. Where sources disagree on data, an official publication is authoritative wherever it gives the number; otherwise the newest analysis prevails; among sources of one kind, the larger sample. The rule decides one visible case. The analysis [2] reads the notice’s class figures 36/36/28 as the rounding of one eleventh per attribute, 4/11, 4/11 and 3/11; but $3/11 = 27.27\%$ does not round to 28%, and the notice is official. The standard therefore uses the notice’s figures, shared equally within a class, and reports the analysis’s reading as a sensitivity (§8).

What the designers did not publish. Three things. The distribution of an increment within its range: the official site gives only averages [7], and a 2018 test of two to three hundred souls [8] states figures that neither a continuous uniform law nor a uniform law on steps of 0.1 reproduces. Whether initial sub-attributes may repeat: the count “2, 3 or 4” presupposes that they do not. And the per-slot drop distribution, outside the object.

3 A FAITHFUL RECONSTRUCTION OF THE DESIGNERS’ PROCESS

3.1 The generative model

Let A be the eleven attributes and $w : A \rightarrow \mathbb{Q}$ the draw weights: $w(a) = 9/100$ for the four 攻击类 and the four 防御类 attributes, and $w(a) = 28/300$ for the three 功能类 attributes. A +15 soul’s sub-attributes are generated as follows.

$N_0 \sim \text{Uniform } \{2, 3, 4\}$
 $I = N_0$ distinct attributes, drawn one by one by w , without replacement
for $t = 1, \dots, 5$:
fewer than four present: $X_t \sim w$ on A , added or strengthened
four present: $X_t \sim \text{Uniform}(\text{present})$, strengthened
an increment of a is $h_a R$, with R i.i.d. on $[4/5, 1]$, mean μ

Three sources of randomness are kept apart: *which attributes exist* (N_0 , I and the adding enhancements), *which attribute each enhancement strengthens* (X_t), and *how large each increment is* (R). The first two are discrete and published; the third is continuous and not.

A soul's value on a in **roll units** is $e_a = S(a)/h_a$, the sum of the R s of a 's increments, and $e_a = 0$ when a is absent. For a set P of attributes, let K_P be the number of increments that land on P .

DEFINITION (The hit law). *The hit law is the distribution of the vector of increment counts per attribute at +15. It is a finite distribution with rational probabilities.*

Computation (Exact propagation). The calibration program computes the hit law by propagating the process above state by state in exact rational arithmetic: a state is the vector of counts, the initial draws are weighted without replacement, and each enhancement is one transition. No simulation and no rounding enters any reference quantity. Table 2 gives the law of K_P for the archetypes of §4.

K	P(K), output and healing	P(K), hit	P(K), speed
0	10.6772%	10.4176%	62.8839%
1	13.8563%	13.7302%	12.0758%
2	20.2318%	20.1178%	15.2167%
3	18.6657%	18.6903%	7.6708%
4	16.1611%	16.2911%	1.9056%
5	11.0711%	11.2136%	0.2349%
6	5.9451%	6.0585%	0.0122%
7	2.5199%	2.5828%	0
8	0.7711%	0.7932%	0
9	0.1006%	0.1050%	0
E[K]	2.907009880	2.929901204	0.743920963

Table 2. The law of the number of useful increments, exact under the official weights.

3.2 The designers were silent on one thing, and it barely matters

Given the hit law, the utility of a set P is a sum of K_P increments, $U_P = R_1 + \dots + R_{K_P}$. The standard needs the expected utility of each archetype, and nothing else from the distribution of R .

PROPOSITION 1 (Mean-only dependence; paper proof). *For every attribute set P , $\mathbb{E}[U_P] = \mu \mathbb{E}[K_P]$, whatever the law of R on $[4/5, 1]$ with mean μ .*

Proof. Condition on the hit law. Given $K_P = k$, U_P is a sum of k increments, each with mean μ and independent of which attribute received it, so $\mathbb{E}[U_P \mid K_P = k] = k\mu$. Averaging over the hit law gives $\mathbb{E}[U_P] = \mu \mathbb{E}[K_P]$. $\square$

Every score in the standard is determined by a soul's own utility and by $\mathbb{E}[U_P]$ (§5), so by Proposition 1 the unpublished law enters a score only through μ . Three independent readings constrain μ : the midpoint of the published range, 0.9; the official site's averages, $2.72/3.0 \approx 0.907$ and $3.65/4.0 \approx 0.913$ [7]; and the 2018 test, whose figures lie between the uniform and the stepped laws [8]. The standard takes $\mu = 9/10$. Its tier rates, which do depend on the full law, are reported under three laws with this mean (§8).

3.3 Why the reconstruction is faithful

Faithfulness here is a list of checkable properties, not a claim about the designers' intent.

1. **Every probability is published.** The weights, the 1/4 rule and the enhancement schedule come from the notice; the ranges from the analysis. The initial count is the one secondary figure, and it is reported as such.

2. **Nothing is fitted.** No parameter is estimated from observed souls, and the tiers are not tuned to a target rate (§6.4).
3. **The propagation is exact.** Every probability of Table 2 is a rational number, printed to four decimals.
4. **The unpublished part is isolated.** By Proposition 1 it enters through one number, and the effect of that number is reported: over the whole published range, $\mu \in [0.8, 1]$, no test vector’s score moves by more than one point from its value at $\mu = 0.9$ (Table 9).
5. **It reproduces what it should.** Under the analysis’s one-eleventh reading the process is invariant under every permutation of the attributes, so four attributes expect exactly $8 \cdot 4/11 = 32/11$ useful increments and one attribute $8/11$, 8 being $\mathbb{E}[N_0] + 5$; the program asserts both of its propagation. The analysis’s theoretical best soul [2] — 18 Crit, 4 CritDmg, 3 AtkPercent and 3 Spd — reaches exactly the attainable maximum (Proposition 2). An independent Monte Carlo simulation of four million souls agrees with every exact tier rate (§8).

4 UTILITY: READING A SOUL AS PLAYERS DO

The reading layer begins here. Each subsection states an adjustment, the player intuition it follows, and what it cannot do.

4.1 Adjustment 1: roll units

Players count lines and rolls: a line that took all five enhancements is 拉满, and 3 Crit and 114 flat HP are both one good increment. The standard measures a sub-attribute in roll units, $e_a = S(a)/h_a$, so that one maximal increment of any attribute is worth 1. Without this, a weighted sum of raw values measures how much flat HP a soul has.

4.2 Adjustment 2: a catalogue of role archetypes

Players judge a soul by what it is for. The standard judges it against a small published catalogue of **archetypes**, each a set A_p of useful sub-attributes with weight 1, a set of main attributes it accepts per slot, and possibly an eligibility predicate on the soul (§7).

Archetype	Useful sub-attributes	Eligible when	Slot 2 mains	Slot 4 mains	Slot 6 mains
output 输出	AtkPercent Crit CritDmg Spd	always	Spd AtkPercent	AtkPercent	Crit CritDmg AtkPercent
hit 命中	EffectHit Spd HpPercent DefPercent	always	Spd HpPercent DefPercent	EffectHit HpPercent DefPercent	HpPercent DefPercent
healing 治疗	HpPercent Crit CritDmg Spd	always	Spd HpPercent	HpPercent	Crit CritDmg HpPercent
speed 速度	Spd	$e_{Spd} \geq 4$	Spd HpPercent DefPercent	EffectHit EffectRes HpPercent DefPercent	HpPercent DefPercent

Table 3. The archetypes of v2. Slots 1, 3 and 5 have one fixed main attribute, which every archetype accepts.

output evaluates general offensive quality; its set is the analysis’s four core output attributes [2]. hit evaluates Effect-Hit and control-oriented quality. healing evaluates the pattern of HP-based healers that can crit; it is output with AtkPercent and HpPercent exchanged, and a general pattern rather than a prescription for any Shikigami. speed evaluates exceptional pure-Speed quality. The last three are authored, and marked so. The archetypes are not exclusive: a soul may be judged by several, and its quality is the best result (§6.5). A soul’s **utility** under archetype p is

$$U_p(s) = \sum_{a \in A_p} e_a(s).$$

Cross-archetype comparability is provided by archetype-specific normalization, not by forcing every archetype to contain the same number of equally useful attributes. Version 1 gave every archetype four attributes, and its third archetype, *resist*, $\{Spd, EffectRes, HpPercent, DefPercent\}$, was chosen partly because it mirrored hit one class member apart, which made its scale provably equal to hit’s. Version 2 removes it. A resist-oriented build is a need of particular Shikigami and sets, which the need-dependent pass judges; EffectRes is useful to no archetype of the quality pass. healing is not given DefPercent to keep the old shape, and

speed is not given filler attributes to reach four. Each archetype carries its own maximum and its own expected soul (§5), and those make the scales comparable.

4.3 Adjustment 3: the main attribute gates, it does not score

Players know that a speed main belongs on slot 2, and that a rare main does not make a good soul by itself. The main attribute decides which archetypes may judge a soul and adds nothing to its utility. A main is worth up to 22 roll units against the sub-attributes' 9 [2]; counting it would let the main decide every score, by its magnitude rather than its use. In particular a Speed main adds nothing to speed's utility: speed reads the Speed sub-attribute alone.

4.4 The attainable maximum, and two structural milestones

PROPOSITION 2 (The attainable maximum; `Archetype.utility_le_nine`, `paperSoul_utility`, `paperSoul_score`). *For every legal soul and four-attribute archetype, $0 \leq U_p(s) \leq 9$; the analysis's theoretical output soul has $U_{\text{output}} = 9$ and scores 100.*

Proof. A line's value is at most its increment count, each increment being at most one roll unit (`Soul.features_le_hits`); a legal soul has at most nine increments, so $U_p \leq \sum_{a \in A_p} \text{hits}(a) \leq 9$. The paper's soul has six maximal increments on Crit and one on each of the other three output attributes, $6 + 1 + 1 + 1 = 9$, and $g(M) = 100$ (Theorem 4). $\square$

PROPOSITION 3 (Six and eight; `Soul.six_hits_of_gt_five`, `Archetype.single_line_le_six`, `beyond_six_needs_another_line`, `nine_useful_of_gt_eight`). *(i) A line above five roll units has taken all five enhancements. (ii) A soul whose useful value lies on one line has $U_p \leq 6$; a soul with $U_p > 6$ has, for every line, another useful line with value. (iii) A soul with $U_p > 8$ has all nine increments on useful lines.*

Proof. (i) $e_a \leq \text{hits}(a) \leq 6$ and $e_a > 5$ force $\text{hits}(a) = 6$. (ii) One line holds at most six increments of at most one roll unit; if every other useful line were zero, U_p would be that line alone, at most 6. (iii) $8 < U_p \leq \sum_{a \in A_p} \text{hits}(a) \leq 9$ forces the useful increment count to be 9. $\square$

Six is therefore the boundary between the best one line can be and quality that needs several lines, and eight the boundary beyond which only the sizes of the increments separate a four-attribute soul from the maximum. §6 builds on both; §7 shows what remains of them for a one-attribute archetype.

5 NORMALIZATION: A COMMON SCALE WITHOUT REWARDING RARITY

5.1 Adjustment 4: three anchors per archetype

A player reads a number against what is usual: *is this better than the souls I normally get?* The standard maps utility to 0–100 by three anchors per archetype, 0 for no useful value, 50 for the archetype's expected +15 soul, and 100 for the archetype's attainable maximum:

$$g_p(u) = \begin{cases} \max(0, 50 u / E_p) & u \leq E_p, \\ \min(100, 50 + 50 (u - E_p) / (M_p - E_p)) & u > E_p, \end{cases}$$

with $E_p = \mu \mathbb{E}[K_{A_p}]$, $M_p = 9$ for the four-attribute archetypes, and $M_{\text{speed}} = 6$. The score of a soul under p is $g_p(U_p(s))$. The anchors are exact rationals; Table 4 gives them.

Parameter set yata-quality-v2. Reference measure: the official class weights, equal within a class. Exact values: `formal/calibration/out/constants.json`.

Milestone	Roll units	Meaning
μ	9/10	mean increment, roll units
M	9	attainable maximum of a four-attribute archetype
u_{SR}	9/2	SR: five useful increments at mean value, 5μ
u_{SSR}	6	SSR: the most one useful line can hold, six increments at maximum

Milestone	Roll units	Meaning
u_SP_floor	63/10	SP's quality floor: seven useful increments at mean value, 7μ
u_UR	8	UR: above $M - 1$
M_speed	6	attainable maximum of speed: one line, six increments at maximum
u_gate	4	speed is a candidate only when $e_Spd \geq 4$
u_UR,speed	5	speed's UR: above $M_speed - 1$

Archetype	E[K]	$E = \mu \cdot E[K]$	M	c_R	c_SR	c_SSR = g(6)	c_SP	t_UR
output	2.907009880	2.616308892	9	50	64.753934	76.502622	78.852360	92.167541
hit	2.929901204	2.636911083	9	50	64.639815	76.426543	78.783889	92.142181
healing	2.907009880	2.616308892	9	50	64.753934	76.502622	78.852360	92.167541

Archetype	E[K]	$E = \mu \cdot E[K]$	M	score at the gate, g(4)	t_UR = g(5)
speed	0.743920963	0.669528867	6	81.239932	90.619966

Table 4. The milestones in roll units, and each archetype's anchors and score thresholds.

THEOREM 4 (Range and anchors; `Anchors.g_nonneg`, `g_le_100`, `g_zero`, `g_E`, `g_M`). *For any anchors $0 < E < M$, $0 \leq g(u) \leq 100$ for all u , and $g(0) = 0$, $g(E) = 50$, $g(M) = 100$.*

Proof. Below E the score is $\max(0, 50u/E)$, which is at least 0 and, for $u \leq E$, at most 50; above E it is the minimum of 100 and a value at least 50. The three values are direct computations. $\square$

THEOREM 5 (Monotonicity; `Anchors.g_mono`, `g_strict`, `g_scale`, `Profile.score_mono`, `score_strict`). *g is nondecreasing, strictly increasing on $[0, M]$, and invariant when the utility and both anchors are scaled together. A soul no worse on every useful attribute scores no lower, and one strictly better on some useful attribute scores strictly higher while within the maximum.*

Proof. Each segment is affine with positive slope, and the lower segment ends at 50 where the upper begins, so g is nondecreasing across the join; the clamps preserve this, and on $[0, M]$ neither is active, which gives strictness. Scaling multiplies the numerator and the denominator of each segment alike. Utility is a sum with nonnegative weights, positive on useful attributes, so improving useful attributes weakly raises it, and strictly when one improves; compose with g . $\square$

5.2 No reward for rarity

THEOREM 6 (No rarity term; `Profile.score_factors`). *Two souls with equal utility under an archetype have equal scores under it, however probable or improbable either soul is.*

Proof. The score is g_p applied to the utility, and g_p depends on the archetype, not on the soul. $\square$

The theorem is short because the design is: probability enters a score only through E_p , a property of the whole archetype, never through how often a particular soul occurs. That is also how a rarer role is kept from gaining. An archetype whose attributes are drawn less often has a lower E_p , and a soul is placed by its distance above that expectation, not by its raw utility. The rejected alternative, a bonus for improbable outcomes, is exactly what Theorem 6 rules out (§10).

5.3 A common scale across archetypes

THEOREM 7 (Relabelling fairness; `expected_relabel`, `score_relabel`, `healing_is_output_relabelled`, `healing_expected_eq_output`). *Let the reference measure be invariant under a permutation π of the attributes. Then an archetype and its image under π have the same expected utility, and with the same maximum, each scores the π -image of a soul exactly as the other scores the soul. The archetype `healing` is the image of `output` under the exchange of `AtkPercent` and `HpPercent`, so under a measure invariant under that exchange the two have the same expected utility.*

Proof. Utility is a sum over attributes, so relabelling both the profile and the features by π leaves it unchanged; reindex the sum by π^{-1} . Invariance of the measure gives a bijection of outcomes that preserves probabilities and relabels features; reindexing the expectation along it shows the two expected utilities equal. The anchors, hence the score functions, are then the same. The image of an equal-weight archetype is the

equal-weight archetype of the image set, and the exchange maps $\{AtkPercent, Crit, CritDmg, Spd\}$ onto $\{HpPercent, Crit, CritDmg, Spd\}$. $\square$

The official weights are invariant under every permutation that keeps attributes in their class. *AtkPercent* is 攻击类 and *HpPercent* 防御类, but both weigh 9%, so the exchange preserves the weights, and healing and output have identical scales; the calibration asserts that their anchors and every tier rate agree. *hit* holds two 功能类 attributes where *output* holds one, so no weight-preserving permutation relates them. Its E_p is under one percent higher, and the residual in tier rates is reported in §8. It is small, and it has the right sign: *hit*'s attributes are drawn slightly more often, so the same utility lies slightly less far above its expectation. Version 1's *resist* was *hit*'s exact partner; with it removed, *hit* has none, and the standard does not claim one.

6 TIERS ON THE FOUR-ATTRIBUTE LADDER

6.1 Adjustment 5: tiers at utility milestones

Players read souls on the game's own rarity ladder, $N < R < SR < SSR < SP < UR$. For the four-attribute archetypes the standard borrows the names and defines each tier by a milestone in roll units, the same for every such archetype, with *R* beginning at the archetype's expected soul E_p :

$$u_{SR} = 5\mu = 4.5 \text{ five useful increments at mean value}$$

$$u_{SSR} = 6 \text{ the most one useful line can hold}$$

$$u_{SP} = 7\mu = 6.3 \text{ SP's quality floor: seven useful increments at mean value}$$

$$u_{UR} = M - 1 = 8 \text{ within one roll unit of the maximum}$$

Each milestone u becomes the score threshold $g_p(u)$ (Table 4). With $spec_p(s)$ true when some useful line holds more than five roll units, the tier is *UR* when the score exceeds $g_p(u_{UR})$; otherwise *SP* when $spec_p(s)$ holds and the score is at least $g_p(u_{SP})$; otherwise *SSR*, *SR*, *R* or *N* as the score reaches $g_p(u_{SSR})$, $g_p(u_{SR})$ or 50.

In players' words, which match the predicates exactly: **N** is below the average soul of its role, **R** at least average, and **SR** about five useful rolls' worth. **SSR** begins where one useful sub-attribute could, in theory, have received all six increments at their maximum; going beyond it needs useful value outside that line. **SP** requires both an extreme specialized line and a higher total quality, so one spectacular line alone remains *SSR*. **UR** is a soul on or next to the best the game allows. These are Yata's scoring decisions, not rarity rules of the game.

THEOREM 8 (The tier function; `milestones_ordered`, `Thresholds.tier_cases`, `tier_UR_iff`, `tier_SP_iff`, `band_ne_SP`, `band_ne_UR`). *The milestones satisfy $u_{SR} < u_{SSR} < u_{SP} < u_{UR}$. For thresholds $c_R < c_{SR} < c_{SSR} < c_{SP} < t_{UR} < 100$ the tier is a function of the score and of specialization, and exactly one of three cases holds: UR iff the score exceeds t_{UR} ; SP iff not UR, specialized and at least c_{SP} ; otherwise the band. A band is never SP or UR.*

Proof. The ordering is arithmetic, $4.5 < 6 < 6.3 < 8$. The tier is a nested conditional on the score and a boolean; a case analysis of the conditional gives exactly one branch, and the band's conditional returns only *N*, *R*, *SR* or *SSR*. $\square$

6.2 Adjustment 6: SSR at one perfect line, SP with a floor of its own

THEOREM 9 (SP needs its floor; `Thresholds.sp_needs_floor`, `sp_is_ssr_quality`, `single_line_not_SP`). *Every SP soul is specialized and at or above c_{SP} , hence above the SSR threshold. If the SP floor lies above $g_p(6)$, a soul whose useful value lies on one line is never SP.*

Proof. The SP branch is taken only when both conditions hold, and $c_{SSR} < c_{SP}$. A soul with one useful line has $U_p \leq 6$ (Proposition 3), so its score is at most $g_p(6)$, below the floor. $\square$

THEOREM 10 (One perfect line is SSR; `perfect_line_at_least_SSR`, `perfect_single_line_is_SSR`). *If the SSR threshold is $g_p(6)$, a soul with six roll units on one useful line ranks at least SSR. If moreover that line is its only useful value, the SP floor is above $g_p(6)$ and the UR boundary is $g_p(8)$, the soul is exactly SSR.*

Proof. The line is at most the whole utility (Archetype.line_le_utility), so $U_p \geq 6$ and the score is at least the SSR threshold, where only SSR, SP and UR occur (Thresholds.at_ssr_floor_high_tier). Theorem 9 excludes SP. Since $U_p \leq 6 < 8$, UR is excluded as well: a soul is UR exactly when $U_p > 8$ (Theorem 11). $\square$

Test vector V04 is this soul, 18 Crit and three useless lines. It is SSR and not SP.

6.3 UR above everything

THEOREM 11 (UR; UR_iff_gt_eight, UR_structure, Thresholds.UR_score_gt, UR_top). *With $M = 9$ and $t_{UR} = g_p(8)$, a soul is UR under p exactly when $U_p > 8$, and then all nine of its increments are useful. Every non-UR soul scores strictly below every UR soul of the same archetype, and every UR soul is strictly first in the exposed order, tier first and then score.*

Proof. g_p is strictly increasing on $[0, 9]$, where legal utilities lie (Proposition 2), so $g_p(U_p) > g_p(8)$ iff $U_p > 8$; Proposition 3 gives the structure. A non-UR score is at most t_{UR} and a UR score exceeds it. A UR soul has rank 5 and every other tier's rank is at most 4. $\square$

6.4 Monotonicity of tiers, and the v1.1 revision

THEOREM 12 (Dominance never lowers a tier; Thresholds.tier_mono, tier_mono_dominance, tier0n_mono_dominance, exposed_mono_dominance). *A soul no worse on every useful attribute of an archetype has, under it, a tier no lower, and at an equal tier a score no lower. This holds for every archetype of the catalogue, speed included.*

Proof. Its score is no lower (Theorem 5), and specialization is preserved upward, since a line above five stays above five. The four-attribute tier function is monotone in the score and in specialization, by case analysis on the order of the thresholds; speed's rule is monotone in its one attribute (Theorem 16). $\square$

The quality of a soul is the best (tier, score) pair over its candidate archetypes. Each archetype's pair is monotone under dominance, and the best of pointwise larger pairs is larger.

THEOREM 13 (Lowering SSR only raises tiers; lower_ssr_floor_never_lowers_tier). *Two threshold sets that agree except that the second has a lower SSR threshold rank every soul at least as high under the second.*

Proof. Case analysis: the UR and SP branches are unchanged; in the band, a lower SSR threshold can only move a score from SR to SSR. $\square$

Version 1 put the SSR edge at $7\mu = 6.3$, the same threshold as the SP floor. That is consistent, but a soul with six maximum increments on one useful line, the best one line can be, was SR. Version 1.1 moved the SSR edge to six and kept the SP floor at 7μ . Scores did not change; by Theorem 13 tiers changed only upward, and only for souls with $6 \leq U_p < 6.3$. Four edges were compared, with the SP floor and the UR boundary fixed:

SP's floor stays $7\mu = 6.3$ and UR's boundary 8 in every row; only SSR's edge moves. The four-attribute archetypes of v2; speed has no SSR edge of this kind.

Candidate	u_SSR	Increment law	Archetype	N	R	SR	SSR	SP	UR	SSR or better
A. v1: 7μ	6.300	continuous uniform on $[0.8, 1]$	output, healing	48.6960%	36.4318%	12.7406%	2.0388%	0.0209%	0.0719%	2.1316%
A. v1: 7μ	6.300	continuous uniform on $[0.8, 1]$	hit	49.3844%	35.4693%	12.9568%	2.0933%	0.0212%	0.0751%	2.1896%
A. v1: 7μ	6.300	7 equal steps on $[0.8, 1]$	output, healing	49.3366%	35.3133%	13.1256%	2.1364%	0.0218%	0.0663%	2.2245%
A. v1: 7μ	6.300	7 equal steps on $[0.8, 1]$	hit	50.6409%	33.7287%	13.3456%	2.1934%	0.0221%	0.0692%	2.2847%
A. v1: 7μ	6.300	2 equal steps on $[0.8, 1]$	output, healing	54.0982%	31.0296%	12.7406%	2.0613%	0.0200%	0.0503%	2.1316%
A. v1: 7μ	6.300	2 equal steps on $[0.8, 1]$	hit	53.6107%	31.2430%	12.9568%	2.1167%	0.0203%	0.0525%	2.1896%
B. v1.1: one perfect line	6.000	continuous uniform on $[0.8, 1]$	output, healing	48.6960%	36.4318%	11.5411%	3.2383%	0.0209%	0.0719%	3.3311%
B. v1.1: one perfect line	6.000	continuous uniform on $[0.8, 1]$	hit	49.3844%	35.4693%	11.7274%	3.3227%	0.0212%	0.0751%	3.4190%

Candidate	u_SSR	Increment law	Archetype	N	R	SR	SSR	SP	UR	SSR or better
B. v1.1: one perfect line	6.000	7 equal steps on [0.8, 1]	output, healing	49.3366%	35.3133%	12.0488%	3.2133%	0.0218%	0.0663%	3.3013%
B. v1.1: one perfect line	6.000	7 equal steps on [0.8, 1]	hit	50.6409%	33.7287%	12.2419%	3.2972%	0.0221%	0.0692%	3.3885%
B. v1.1: one perfect line	6.000	2 equal steps on [0.8, 1]	output, healing	54.0982%	31.0296%	11.5452%	3.2567%	0.0200%	0.0503%	3.3270%
B. v1.1: one perfect line	6.000	2 equal steps on [0.8, 1]	hit	53.6107%	31.2430%	11.7321%	3.3414%	0.0203%	0.0525%	3.4142%
C. 6 μ	5.400	continuous uniform on [0.8, 1]	output, healing	48.6960%	36.4318%	8.5081%	6.2713%	0.0209%	0.0719%	6.3641%
C. 6 μ	5.400	continuous uniform on [0.8, 1]	hit	49.3844%	35.4693%	8.6361%	6.4140%	0.0212%	0.0751%	6.5103%
C. 6 μ	5.400	7 equal steps on [0.8, 1]	output, healing	49.3366%	35.3133%	8.7502%	6.5118%	0.0218%	0.0663%	6.5999%
C. 6 μ	5.400	7 equal steps on [0.8, 1]	hit	50.6409%	33.7287%	8.8799%	6.6592%	0.0221%	0.0692%	6.7505%
C. 6 μ	5.400	2 equal steps on [0.8, 1]	output, healing	54.0982%	31.0296%	7.5792%	7.2227%	0.0200%	0.0503%	7.2930%
C. 6 μ	5.400	2 equal steps on [0.8, 1]	hit	53.6107%	31.2430%	7.6894%	7.3841%	0.0203%	0.0525%	7.4569%
D. SSR or better at 5% (uniform, output)	5.488	continuous uniform on [0.8, 1]	output, healing	48.6960%	36.4318%	9.8667%	4.9127%	0.0209%	0.0719%	5.0055%
D. SSR or better at 5% (uniform, output)	5.488	continuous uniform on [0.8, 1]	hit	49.3844%	35.4693%	10.0206%	5.0294%	0.0212%	0.0751%	5.1257%
D. SSR or better at 5% (uniform, output)	5.488	7 equal steps on [0.8, 1]	output, healing	49.3366%	35.3133%	10.1216%	5.1404%	0.0218%	0.0663%	5.2285%
D. SSR or better at 5% (uniform, output)	5.488	7 equal steps on [0.8, 1]	hit	50.6409%	33.7287%	10.2774%	5.2616%	0.0221%	0.0692%	5.3529%
D. SSR or better at 5% (uniform, output)	5.488	2 equal steps on [0.8, 1]	output, healing	54.0982%	31.0296%	9.4370%	5.3649%	0.0200%	0.0503%	5.4352%
D. SSR or better at 5% (uniform, output)	5.488	2 equal steps on [0.8, 1]	hit	53.6107%	31.2430%	9.5827%	5.4908%	0.0203%	0.0525%	5.5636%

Table 5. SSR edge candidates under every increment law, for the four-attribute archetypes of v2. Only the SSR edge moves.

The edge of v1, 7 μ , leaves the best single line at SR. The edge 6 μ = 5.4 falls in the middle of what six increments at mean value produce, so its rate moves most with the unpublished increment law, and it admits souls whose best line is well below perfect. An edge set so that 5% of souls are SSR or better has no structural meaning and moves with the assumed law. Six has a structural meaning, keeps SSR rare — about one soul in thirty is SSR or better under one archetype — and depends least on what the designers did not publish: a soul reaches it only with a seventh useful increment, or with exactly six useful increments all at maximum, so its rate follows the exact hit law.

6.5 The exposed order, and the quality of a soul

The exposed order is tier first, then score. UR is first in both orders within an archetype (Theorem 11). SP is above SSR in tier order, but an SSR soul may outscore an SP soul by up to $t_{UR} - c_{SP}$: SP marks a kind of excellence, a fully specialized line on an already strong soul, not a higher number. A soul's quality is its best (tier, score) under its candidate archetypes. A soul with several candidates has several chances, which raises its tier distribution (Table 8). This is fit, not rarity: the commonest mains, the flat mains of slots 1, 3 and 5, share the advantage with the rare speed main.

7 A ONE-ATTRIBUTE ARCHETYPE: SPEED

A soul whose value is one long Speed line is kept for its Speed alone. Judged by the four-attribute archetypes, it reaches at most six of nine roll units (Proposition 3) and reads as an ordinary soul: 12.0 Speed beside two useless lines is SR under hit (V19). The speed archetype reads the Speed line by itself, and only when it is long enough to mean something.

7.1 Adjustment 7: a gate at four roll units

speed is a **candidate** for a soul only when $e_{Spd} \geq 4$, 12 displayed Speed at six stars; every other archetype is always eligible. The gate reads the result, the line's value, not the number of increments inferred from it: a line of four maximal increments and a line of five minimal ones, both exactly four roll units, are judged alike (V19, V21). Under the published range the two readings nearly coincide — five increments always reach four roll units, and four increments reach it only at their maximum — but a gate stated on inferred history would let four low increments outrank fewer high ones for the wrong reason. Four roll units is Yata's choice, not a rule of the game.

PROPOSITION 14 (Candidates; Arch.coverage, candidate_exists_iff, speed_not_candidate, speed_gate_mono). *Every legal main attribute of every slot is accepted by some archetype. A legal soul has a candidate archetype unless its main is EffectRes on slot 4 and $e_{Spd} < 4$. Below the gate speed is not a candidate, and raising Speed never removes it from the candidates.*

Proof. Coverage and the characterization are case analyses over the six slots and their legal mains: every main but slot 4's EffectRes is accepted by a four-attribute archetype, which is always eligible, and EffectRes on slot 4 is accepted by speed alone. The gate is an inequality on e_{Spd} , preserved when e_{Spd} grows. $\square$

With resist removed, a slot 4 EffectRes soul whose Speed is below the gate has no candidate. It is **unrated** in the quality pass: no score and no tier, and its value is a question for the need-dependent pass. Version 1 required every soul to receive a quality score; that requirement was a consequence of resist, and it is not kept by re-adding it. Below the gate, speed is absent, not scored low: a low speed score competing in the best-of selection would say nothing the other archetypes do not.

7.2 The gate is not a condition on the reference

speed's anchors are $E_{speed} = \mu \mathbb{E}[K_{Spd}]$ over all +15 souls, and $M_{speed} = 6$. The alternative conditions the reference on the gate, taking $E' = \mathbb{E}[e_{Spd} \mid e_{Spd} \geq 4]$. The two answer different questions. The gate decides when the Speed reading is shown; the normalization says where a Speed line stands in the Speed dimension of the game's souls. Conditioning would place 60% of eligible souls below 50 and score the gate itself at 44 — calling a soul in the top quarter percent of Speed below the expected soul — while the unconditional reference scores the gate at 81.24, above every four-attribute SSR threshold, and no eligible soul below it (Table 6). The standard uses the unconditional reference, and checks that it produces no absurd score at the boundary.

7.3 Tiers for one attribute

The four-attribute ladder does not transfer. Its SSR milestone, six roll units, is speed's maximum; its UR boundary, eight, is unreachable; and every eligible soul already lies above its R and SR milestones. Its specialization predicate, some useful line above five roll units, is exactly the statement that the soul is within one roll unit of speed's maximum. The standard therefore gives speed a rule of its own: an eligible soul is SSR, and UR when $e_{Spd} > M_{speed} - 1 = 5$ — the same frontier principle as the four-attribute UR. speed has no SP: an SP rule built on the generic predicate would never fire below UR, and the standard does not invent a second condition to fill the tier.

THEOREM 15 (The speed scale; speed_utility, speed_utility_bounds, speed_score_mono, speed_score_strict, maxSpeedSoul_score). *$U_{speed}(s) = e_{Spd}$ and $0 \leq U_{speed} \leq 6$. Raising Speed never lowers the speed score; the score is strictly increasing over $[0, 6]$; and the theoretical maximum Speed line, six increments at maximum, scores 100.*

Proof. speed values one attribute with weight 1, so its utility is that attribute’s value, at most its six increments of at most one roll unit each. Monotonicity and strictness are Theorem 5 for this profile with $M = 6$, and $g(M) = 100$ (Theorem 4). $\square$

THEOREM 16 (Speed tiers; speedTier_cases, speedTier_high, speedTier_ne_SP, speedTier_mono, speed_specialized_iff_UR, speed_UR_six_hits, speed_candidate_rank, speed_milestones). *speed’s tier is a function of e_{Spd} , and exactly one of two cases holds: UR with $e_{Spd} > 5$, or SSR with $e_{Spd} \leq 5$. It is never N, R, SR or SP, it is monotone in e_{Spd} , and every candidate is at least SSR. The generic specialization predicate holds under speed exactly when the soul is UR, and a UR soul put all six increments on Speed. The gate lies below the UR boundary, which is one roll unit below the maximum.*

Proof. The rule is one conditional on e_{Spd} . Specialization under a one-attribute archetype is $e_{Spd} > 5$, the UR condition; Proposition 3 (i) gives six increments. The milestones are arithmetic, $4 < 5 = 6 - 1$. $\square$

Of all +15 souls, under the reference measure; speed has no N, R, SR or SP.

Increment law	P(eligible)	SSR	UR	eligible: 1 in
continuous uniform on [0.8, 1]	0.2471%	0.2349%	0.0122%	405
7 equal steps on [0.8, 1]	0.2479%	0.2358%	0.0121%	403
2 equal steps on [0.8, 1]	0.3662%	0.3554%	0.0109%	273

Of eligible souls: where e_{Spd} lies, the speed score and tier there, and how many increments the Speed line has.

Increment law	[4, 4.5]	[4.5, 5]	(5, 5.5]	(5.5, 6]	4 increments	5 increments	6 increments
continuous uniform on [0.8, 1]	47.5306%	47.5374%	3.7245%	1.2075%	0.0000%	95.0612%	4.9388%
7 equal steps on [0.8, 1]	43.6082%	51.5074%	3.6927%	1.1917%	0.3201%	94.7568%	4.9230%
2 equal steps on [0.8, 1]	64.5940%	32.4378%	1.8226%	1.1456%	32.5207%	64.1466%	3.3327%
speed score	81.24 to 85.93	85.93 to 90.62	90.62 to 95.31	95.31 to 100.00			
tier	SSR	SSR	UR	UR			

The reference of speed’s normalization, uniform increments: A, the standard, is unconditional; B would condition it on the gate.

Reference	E	score at $e_{Spd} = 4$	4.5	5	6	eligible souls scored below 50
A. unconditional (the standard)	0.6695	81.24	85.93	90.62	100.00	0.0000%
B. conditioned on $e_{Spd} \geq 4$	4.5444	44.01	49.51	65.65	100.00	59.9700%

Table 6. speed under the reference measure: how often a +15 soul is eligible, where the eligible souls lie, and the two candidate references.

About one +15 soul in four hundred is a speed candidate; nineteen in twenty of those have five increments on Speed, and the rest six. One soul in about eight thousand is speed UR, rarer than the four-attribute UR, as a single line near its maximum should be. Across archetypes the score thresholds differ: a speed UR soul scores at least $g_{speed}(5) \approx 90.6$, below the four-attribute $t_{UR} \approx 92.2$, and it still ranks above every SSR and SP soul, because the exposed order is tier first. The gate makes speed a candidate, not the winner: a speed-eligible soul whose four-attribute reading is higher is best fit under that archetype.

8 CALIBRATION

Every table in this section is generated by formal/calibration/ and checked against the program by the repository’s preflight (§9).

8.1 Tier rates

Increment law	Archetype	N	R	SR	SSR	SP	UR
continuous uniform on [0.8, 1]	output, healing	48.6960%	36.4318%	11.5411%	3.2383%	0.0209%	0.0719%
continuous uniform on [0.8, 1]	hit	49.3844%	35.4693%	11.7274%	3.3227%	0.0212%	0.0751%
7 equal steps on [0.8, 1]	output, healing	49.3366%	35.3133%	12.0488%	3.2133%	0.0218%	0.0663%
7 equal steps on [0.8, 1]	hit	50.6409%	33.7287%	12.2419%	3.2972%	0.0221%	0.0692%
2 equal steps on [0.8, 1]	output, healing	54.0982%	31.0296%	11.5452%	3.2567%	0.0200%	0.0503%
2 equal steps on [0.8, 1]	hit	53.6107%	31.2430%	11.7321%	3.3414%	0.0203%	0.0525%

Table 7. Tier rates of one four-attribute archetype under the official weights, for three increment laws with mean 0.9; output and healing are identical.

About half of all +15 souls are below the average soul of a given four-attribute archetype, one in nine is SR, one in thirty is SSR or better, and about one in fourteen hundred is UR. SP is rarer than UR: a line must take all five enhancements, which happens to a given line with probability $(1/4)^5$ even when all four lines exist. The ladder orders quality, not rarity.

4000000 souls per row, seed 0x594154417631, uniform increments, the reference measure. Each cell: rate (standard error), and its z-score against the exact rate. For speed, “not eligible” is $e_Spd < 4$.

Row	N	R	SR	SSR	SP	UR	not eligible
exact, output	48.6960%	36.4318%	11.5411%	3.2383%	0.0209%	0.0719%	0.0000%
Monte Carlo, output	48.7182% (0.0250%) z=+0.89	36.4113% (0.0241%) z=-0.85	11.5315% (0.0160%) z=-0.60	3.2486% (0.0089%) z=+1.17	0.0199% (0.0007%) z=-1.28	0.0705% (0.0013%) z=-1.08	0.0000% (0.0000%)
exact, hit	49.3844%	35.4693%	11.7274%	3.3227%	0.0212%	0.0751%	0.0000%
Monte Carlo, hit	49.3473% (0.0250%) z=-1.48	35.5240% (0.0239%) z=+2.29	11.7048% (0.0161%) z=-1.40	3.3295% (0.0090%) z=+0.76	0.0200% (0.0007%) z=-1.64	0.0744% (0.0014%) z=-0.47	0.0000% (0.0000%)
exact, healing	48.6960%	36.4318%	11.5411%	3.2383%	0.0209%	0.0719%	0.0000%
Monte Carlo, healing	48.7135% (0.0250%) z=+0.70	36.4206% (0.0241%) z=-0.46	11.5309% (0.0160%) z=-0.64	3.2430% (0.0089%) z=+0.53	0.0192% (0.0007%) z=-2.25	0.0727% (0.0013%) z=+0.56	0.0000% (0.0000%)
exact, speed	0.0000%	0.0000%	0.0000%	0.2349%	0.0000%	0.0122%	99.7529%
Monte Carlo, speed	0.0000% (0.0000%)	0.0000% (0.0000%)	0.0000% (0.0000%)	0.2338% (0.0024%) z=-0.48	0.0000% (0.0000%)	0.0118% (0.0005%) z=-0.75	99.7544% (0.0025%) z=+0.64

Largest $|z|$ of a Monte Carlo rate against the exact rate: 2.29.

Quality over the candidate archetypes	Accepting	N	R	SR	SSR	SP	UR	unrated
slots 1, 3, 5; Spd on 2	output, hit, healing, speed	19.3472% (0.0198%)	48.0466% (0.0250%)	23.8363% (0.0213%)	8.5068% (0.0139%)	0.0340% (0.0009%)	0.2291% (0.0024%)	0.0000% (0.0000%)
AtkPercent on 2, 4, 6	output	48.7182% (0.0250%)	36.4113% (0.0241%)	11.5315% (0.0160%)	3.2486% (0.0089%)	0.0199% (0.0007%)	0.0705% (0.0013%)	0.0000% (0.0000%)
Crit, CritDmg on 6	output, healing	36.4321% (0.0241%)	41.7664% (0.0247%)	16.2661% (0.0185%)	5.3640% (0.0113%)	0.0283% (0.0008%)	0.1431% (0.0019%)	0.0000% (0.0000%)
HpPercent on 2, 4, 6	hit, healing, speed	28.1760% (0.0225%)	45.8030% (0.0249%)	19.4226% (0.0198%)	6.4132% (0.0122%)	0.0264% (0.0008%)	0.1588% (0.0020%)	0.0000% (0.0000%)
EffectHit on 4; DefPercent on 2, 4, 6	hit, speed	49.3473% (0.0250%)	35.4905% (0.0239%)	11.5736% (0.0160%)	3.4878% (0.0092%)	0.0147% (0.0006%)	0.0861% (0.0015%)	0.0000% (0.0000%)
EffectRes on 4	speed	0.0000% (0.0000%)	0.0000% (0.0000%)	0.0000% (0.0000%)	0.2338% (0.0024%)	0.0000% (0.0000%)	0.0118% (0.0005%)	99.7544% (0.0025%)

Table 8. A Monte Carlo check of the exact rates, and the quality of a soul over its candidate archetypes, for every distinct set of accepting archetypes.

The four archetypes are scored on the same souls, so their Monte Carlo counts are correlated and form one sample, not four. A conditional simulation of the values given the exact hit law, a simulation of the hit process alone, and a separate simulation of twenty million souls agreed with the exact SP rate of version 1.1, which v2 leaves unchanged; these were development checks and are not in the repository.

8.2 Sensitivity to what was not published

μ	E, output and healing	E, hit	E, speed	score V03	score V05	score V11	score V22
0.8 (all rolls minimal)	2.3256	2.3439	0.5951	80.52	87.26	98.15	88.90
0.9 (the standard: midpoint)	2.6163	2.6369	0.6695	79.64	86.68	98.12	88.74
0.9067 (官网 2.72/3.0)	2.6357	2.6564	0.6745	79.57	86.64	98.12	88.73
0.9125 (官网 3.65/4.0)	2.6526	2.6735	0.6788	79.52	86.61	98.12	88.72
1.0 (all rolls maximal)	2.9070	2.9299	0.7439	78.66	86.05	98.10	88.58

Table 9. Anchors, and the scores of four test vectors under their best-fit archetypes, under other means of the increment.

Measure	Archetype	E	N	R	SR	SSR	SP	UR
36/36/28 per class (official notice; the reference)	output	2.6163	48.6960%	36.4318%	11.5411%	3.2383%	0.0209%	0.0719%

Measure	Archetype	E	N	R	SR	SSR	SP	UR
36/36/28 per class (official notice; the reference)	hit	2.6369	49.3844%	35.4693%	11.7274%	3.3227%	0.0212%	0.0751%
36/36/28 per class (official notice; the reference)	healing	2.6163	48.6960%	36.4318%	11.5411%	3.2383%	0.0209%	0.0719%
1/11 per attribute (Hu 2026 reading)	output	2.6182	48.6513%	36.4509%	11.5584%	3.2464%	0.0209%	0.0722%
1/11 per attribute (Hu 2026 reading)	hit	2.6182	49.8328%	35.2693%	11.5584%	3.2464%	0.0209%	0.0722%
1/11 per attribute (Hu 2026 reading)	healing	2.6182	48.6513%	36.4509%	11.5584%	3.2464%	0.0209%	0.0722%
official weights; initial count 1/2, 1/3, 1/6	output	2.5073	50.8994%	36.4873%	10.3522%	2.2146%	0.0105%	0.0360%
official weights; initial count 1/2, 1/3, 1/6	hit	2.5273	51.6120%	35.5242%	10.5395%	2.2761%	0.0107%	0.0375%
official weights; initial count 1/2, 1/3, 1/6	healing	2.5073	50.8994%	36.4873%	10.3522%	2.2146%	0.0105%	0.0360%

Measure	E, speed	P(eligible)	SSR	UR
36/36/28 per class (official notice; the reference)	0.6695	0.2471%	0.2349%	0.0122%
1/11 per attribute (Hu 2026 reading)	0.6545	0.2410%	0.2291%	0.0119%
official weights; initial count 1/2, 1/3, 1/6	0.6419	0.1516%	0.1454%	0.0062%

Table 10. Tier rates and speed’s eligibility under the official weights, the analysis’s one-eleventh reading, and a skewed initial count, all with v_2 ’s thresholds.

The two readings of the weights differ in E_p by under one percent and in any tier rate by under half a percentage point. Under the one-eleventh reading all three four-attribute archetypes would share one expected utility.

8.3 Comparing normalizations

1000000 souls, seed 0x4E4F524D, uniform increments; the same souls for every profile. Each cell: share of souls scoring at or above the output archetype’s c_R / c_SR / c_SSR / above its t_UR , and the mean score.

Normalization	Profile	$\geq c_R$	$\geq c_SR$	$\geq c_SSR$	$> t_UR$	mean
A. bound: 100-U/M	speed only (1 attribute)	2.153%	0.247%	0.070%	0.0019%	11.16
A. bound: 100-U/M	crit pair (2 attributes)	8.774%	2.035%	0.616%	0.0197%	18.54
A. bound: 100-U/M	output archetype (4 attributes)	14.883%	3.387%	0.872%	0.0134%	29.08
C. mean ratio: min(100, 50-U/E)	speed only (1 attribute)	37.183%	33.089%	25.022%	25.0223%	33.19
C. mean ratio: min(100, 50-U/E)	crit pair (2 attributes)	44.801%	43.026%	23.312%	23.2598%	43.90
C. mean ratio: min(100, 50-U/E)	output archetype (4 attributes)	51.295%	36.025%	20.394%	9.4334%	48.95
H. anchored (the standard)	speed only (1 attribute)	37.183%	9.821%	1.784%	0.0119%	22.54
H. anchored (the standard)	crit pair (2 attributes)	44.801%	10.053%	3.470%	0.1019%	33.12
H. anchored (the standard)	output archetype (4 attributes)	51.295%	14.883%	3.324%	0.0748%	42.29
B. percentile (mid-rank)	speed only (1 attribute)	37.183%	35.246%	23.497%	7.8325%	50.00
B. percentile (mid-rank)	crit pair (2 attributes)	50.000%	35.246%	23.497%	7.8325%	50.00
B. percentile (mid-rank)	output archetype (4 attributes)	50.000%	35.246%	23.497%	7.8325%	50.00

Table 11. Four normalizations on profiles of one, two and four useful attributes, on the same simulated souls, against the output archetype’s score thresholds.

Soul	U	anchored score (any law with mean 0.9)	percentile, uniform	percentile, 7 steps	percentile, 2 steps	P(equal or better), uniform
V02	2.7000	50.66	54.10	54.10	54.10	45.9018% (1 in 2)
V03	6.4000	79.64	98.47	98.40	98.21	1.5276% (1 in 65)
V05	7.3000	86.68	99.69	99.67	99.62	0.3115% (1 in 321)
V10	8.0000	92.17	99.93	99.93	99.94	0.0719% (1 in 1390)

Table 12. The anchored score, the percentile under three increment laws, and the probability of an equal or better soul.

Table 11 is also why speed has a tier rule of its own. On the four-attribute ladder, a one-attribute profile’s tails differ from a four-attribute profile’s under every normalization but the percentile, and the percentile compresses the top. speed keeps the anchored scale, and replaces the ladder by a gate and a frontier.

9 MECHANIZATION AND REPRODUCIBILITY

The Lean development in `formal/lean/` has seven modules, pinned to Lean 4.34.0 and Mathlib v4.34.0 [4, 5]. It models exactly what the theorems need: attributes, slots, a soul as its increments per attribute, profiles and archetypes of any size, eligibility and candidates, anchors, thresholds and tiers, speed’s tier rule, and a finite reference measure. It builds with no sorry; the theorems depend only on propositional extensionality, choice and quotient soundness, the axioms of Mathlib’s classical logic. The anchors are parameters in Lean satisfying $0 < E < M$; their values, and every rate, come from the calibration program, which is tested and not proved.

The calibration program in `formal/calibration/` is a standalone Rust program. It computes the hit law exactly with arbitrary-precision rationals; the tier rates of the continuous uniform law exactly through the Irwin–Hall distribution [1, 3], with one numerical integral for the SP rate; the stepped laws by exact convolution; and the Monte Carlo check with a fixed seed. It writes the generated blocks of this paper and of the specification, and asserts the facts the text states in words: the milestones ordered and equal to the Lean definitions, 32/11 and 8/11 under the one-eleventh reading, output and healing equal, the Monte Carlo groups equal to the catalogue’s acceptance, one perfect line below the SP floor, and what each golden vector exists to show. Preflight runs `lake build` and the program’s check; a changed definition not reflected in both fails the build.

```
cd formal/lean && lake exe cache get && lake build
cargo run --manifest-path formal/calibration/Cargo.toml --release -- check \
docs/spec/quality-model.md papers/quality-model-v2/paper.md
```

10 REJECTED ALTERNATIVES

Each alternative below was computed or proved against before it was rejected.

1. **A bonus for rarity**, $U + \lambda(-\log P(U' \geq U))$. Two souls of equal utility would score differently by how often the game produces such a soul; Theorem 6 rules this out.
2. **Percentile normalization**. Exactly fair by construction and, here, robust to the increment law; but it compresses the top, placing SSR, SP and near-UR souls within about one and a half points (Table 12). It is kept as an explanation, never as the score.
3. **Utility over the maximum**. The expected soul would score under 30, and a one-attribute profile’s scores a third of a four-attribute profile’s (Table 11).
4. **Utility over the expectation**. Unbounded; a quarter of speed-only souls saturate at 100 (Table 11).
5. **UR as distance to the Pareto frontier**. With weights (2, 1, 1, 1), the frontier point with six maximal increments on a weight-1 line has utility 10, while a non-frontier soul with 5.5 roll units on the weight-2 line has 13.9, so an SP would outscore a UR. For equal weights the utility gap is the distance to the frontier.
6. **SP as “one attribute took every enhancement”**. A perfect line beside three useless ones would be SP; SP needs a floor of its own.
7. **SSR at 7μ** , version 1. Superseded: the best single line stayed SR (§6.4).
8. **resist kept for symmetry**, version 1’s third archetype. Removed: its reason was the shape of the catalogue, and EffectRes is a need-dependent question. **healing with DefPercent** and **speed with filler attributes** were rejected for the same reason.
9. **speed normalized against the souls that pass its gate**. Most eligible souls would score below 50 (§7.2).
10. **speed on the four-attribute ladder**, or **one-attribute profiles on the ladder in general**, whose tails differ from four-attribute profiles (Table 11). Rejected; speed has a gate and a frontier instead (§7.3). **An SP for speed from an invented second condition** was rejected as a tier with no meaning of its own.
11. **The main attribute in utility** (§4.3), and **depth as the concentration of all value**, which would reward concentration on a useless attribute.

11 DISCUSSION AND LIMITS

What faithfulness covers. The measure is the published process, and the reading is the standard’s. The standard is faithful in that the reading never alters the measure (Theorem 6), and the measure takes nothing that was not published, except where §2 says so. The eligibility gate is part of the reading: it decides which readings are shown and leaves the measure, and every anchor computed from it, unchanged.

Most souls are R. Within one four-attribute archetype about half of all souls are N; over the candidate archetypes, the plurality of souls with a flexible main is R (Table 8). This follows from taking the best of several archetypes, not from where the R edge lies, and the standard keeps it: R reads as “at least average for some role the soul can serve”.

Weights are equal. Every useful attribute has weight 1. The maintainer expects `EffectHit` and `Spd` to weigh more in `hit` in a future weighted model. That is a separate calibration problem; this version isolates the catalogue from it, and `Rejected` alternative 5 records the constraint weights put on `UR`.

The formal and numerical claims are further bounded by the following.

- *Six stars only.* Lower-star ranges are not published.
- *The increment law is unpublished.* Scores depend on it only through μ ; tier rates are reported under three laws.
- *The catalogue is authored.* `hit`, healing and speed, their mains, and the gate await review; a changed catalogue is a new model version.
- *Interactions are out of the quality pass.* Crit's cap, speed thresholds and set effects belong to the need-dependent pass.
- *Candidates use current values.* A soul below +15 whose Speed may later pass the gate is not yet judged by speed; its growth is reported under its current best fit.
- *The calibration is tested, not proved.* It is checked against Monte Carlo and against the Lean definitions of the milestones.
- *Nothing about players.* The adjustments follow how players read souls; whether they succeed is an empirical question.

12 CONCLUSION

The standard is interesting not because it produces a number, but because it can say which parts of that number belong to the game and which to the standard. The game's part is the published enhancement process, reconstructed exactly, with the one unpublished quantity entering only through its mean. The standard's part is a short list of declared adjustments — roll units, a catalogue of archetypes of different sizes, main-attribute gating and a Speed gate, three anchors per archetype, and tiers at utility milestones — each chosen to read as players read souls, and each proved not to feed rarity back into the score. That the adjustments preserve bounds, monotonicity and dominance, that healing and output share one scale because the rules treat their attributes alike, that one perfect line is exactly SSR, that `UR` is above everything of its archetype, and that speed is judged only above its gate and only by a rule that means something for one line, are theorems; that the rates are what the published rules imply is a computation anyone can repeat.

APPENDIX A — THEOREM INDEX

Every name is a Lean declaration in `formal/lean/` of `KCN-judu/yata`, in the namespace `Yata`.

Paper	Lean names	File
Prop 1	paper proof; computed by the calibration's <code>mean_useful</code>	—
Prop 2	<code>Archetype.utility_le_nine</code> , <code>utility_le_useful_hits</code> , <code>paperSoul_utility</code> , <code>paperSoul_score</code>	Archetype
Prop 3	<code>Soul.six_hits_of_gt_five</code> , <code>Archetype.single_line_le_six</code> , <code>beyond_six_needs_another_line</code> , <code>nine_useful_of_gt_eight</code> , <code>line_le_utility</code>	Soul, Archetype
Thm 4	<code>Anchors.g_nonneg</code> , <code>g_le_100</code> , <code>g_zero</code> , <code>g_E</code> , <code>g_M</code>	Score
Thm 5	<code>Anchors.g_mono</code> , <code>g_strict</code> , <code>g_scale</code> ; <code>Profile.utility_mono</code> , <code>utility_strict</code> , <code>score_mono</code> , <code>score_strict</code>	Score, Soul
Thm 6	<code>Profile.score_factors</code>	Soul
Thm 7	<code>expected_relabel</code> , <code>score_relabel</code> , <code>utility_relabel</code> ; <code>archetype_relabel</code> , <code>healing_is_output_relabelled</code> , <code>healing_expected_eq_output</code>	Fairness, Quality
Thm 8	<code>milestones_ordered</code> ; <code>Thresholds.tier_cases</code> , <code>tier_UR_iff</code> , <code>tier_SP_iff</code> , <code>band_ne_SP</code> , <code>band_ne_UR</code> ; <code>Tier.ladder</code>	Quality, Tier
Thm 9	<code>Thresholds.sp_needs_floor</code> , <code>sp_is_ssr_quality</code> ; <code>single_line_not_SP</code>	Tier, Quality
Thm 10	<code>perfect_line_at_least_SSR</code> , <code>perfect_single_line_is_SSR</code> ; <code>Thresholds.at_ssr_floor_high_tier</code> , <code>ssr_floor_rank</code>	Quality, Tier
Thm 11	<code>UR_iff_gt_eight</code> , <code>UR_structure</code> , <code>paperSoul_UR</code> ; <code>Thresholds.UR_score_gt</code> , <code>UR_top</code>	Quality, Tier
Thm 12	<code>Thresholds.tier_mono</code> ; <code>tier_mono_dominance</code> , <code>tierOn_mono_dominance</code> , <code>tierOn_broad</code> , <code>exposed_mono_dominance</code>	Tier, Quality
Thm 13	<code>lower_ssr_floor_never_lowers_tier</code>	Tier
Prop 14	<code>Arch.coverage</code> , <code>Arch.accepts_legal</code> , <code>Arch.card_broad</code> , <code>Arch.card_speed</code> , <code>candidate_exists_iff</code> , <code>speed_not_candidate</code> , <code>speed_candidate_iff</code> , <code>speed_gate_mono</code> , <code>gate_broad</code>	Quality
Thm 15	<code>speed_utility</code> , <code>speed_utility_bounds</code> , <code>speed_score_mono</code> , <code>speed_score_strict</code> , <code>maxSpeedSoul_speed</code> , <code>maxSpeedSoul_score</code>	Quality
Thm 16	<code>speedTier_cases</code> , <code>speedTier_high</code> , <code>speedTier_ne_SP</code> , <code>speedTier_mono</code> , <code>speed_specialized_iff_UR</code> , <code>speed_UR_six_hits</code> , <code>speed_candidate_rank</code> , <code>speed_milestones</code>	Quality

APPENDIX B — GOLDEN TEST VECTORS

Stored values are in display units, with each line's increment count in parentheses; the full results per archetype are in `formal/calibration/out/test-vectors.json`.

Id	Soul	Slot, main, level	Best fit	U	Score	Specialized	Tier	Depth	Breadth	Growth	Every candidate
V01	AtkFlat 72.0 (3); HpFlat 205.0 (2); DefFlat 13.5 (3); EffectRes 3.6 (1)	1, AtkFlat, +15	output	0.0000	0.00	no	N	0.0	0.0	—	output N 0.00; hit N 0.00; healing N 0.00
V02	Crit 5.4 (2); AtkPercent 2.7 (1); HpFlat 300.0 (3); DefFlat 13.8 (3)	3, DefFlat, +15	output	2.7000	50.66	no	R	30.0	26.7	—	output R 50.66; healing N 34.40; hit N 0.00
V03	Crit 8.4 (3); Spd 5.4 (2); AtkPercent 5.4 (2); HpFlat 205.0 (2)	6, CritDmg, +15	output	6.4000	79.64	no	SSR	46.7	62.0	—	output SSR 79.64; healing SR 65.54
V04	Crit 18.0 (6); HpFlat 114.0 (1); DefFlat 5.0 (1); AtkFlat 27.0 (1)	1, AtkFlat, +15	output	6.0000	76.50	yes	SSR	100.0	0.0	—	output SSR 76.50; healing SSR 76.50; hit N 0.00
V05	Crit 16.5 (6); CritDmg 3.6 (1); AtkPercent 2.7 (1); HpFlat 100.0 (1)	5, HpFlat, +15	output	7.3000	86.68	yes	SP	91.7	22.4	—	output SP 86.68; healing SP 79.64; hit N 0.00
V06	Crit 15.0 (5); CritDmg 3.6 (1); AtkPercent 2.7 (1); HpFlat 200.0 (2)	5, HpFlat, +15	output	6.8000	82.77	no	SSR	83.3	24.6	—	output SSR 82.77; healing SR 75.72; hit N 0.00
V07	Crit 15.3 (6); AtkPercent 3.0 (1); DefFlat 4.5 (1); HpFlat 100.0 (1)	1, AtkFlat, +15	output	6.1000	77.29	yes	SSR	85.0	12.6	—	output SSR 77.29; healing SR 69.45; hit N 0.00
V08	Crit 18.0 (6); CritDmg 4.0 (1); AtkPercent 3.0 (1); Spd 3.0 (1)	6, CritDmg, +15	output	9.0000	100.00	yes	UR	100.0	35.9	—	output UR 100.00; healing SP 92.17
V09	Crit 8.7 (3); CritDmg 7.8 (2); AtkPercent 5.8 (2); Spd 5.8 (2)	2, Spd, +15	output	8.7167	97.78	no	UR	48.3	95.3	—	output UR 97.78; healing SSR 82.64; hit N 36.66
V10	Crit 18.0 (6); CritDmg 4.0 (1); AtkPercent 3.0 (1); HpFlat 114.0 (1)	6, CritDmg, +15	output	8.0000	92.17	yes	SP	100.0	22.8	—	output SP 92.17; healing SP 84.34
V11	Spd 17.4 (6); EffectHit 3.6 (1); HpPercent 2.7 (1); DefFlat 4.5 (1)	2, Spd, +15	speed	5.8000	98.12	yes	UR	96.7	n/a	—	speed UR 98.12; hit SP 89.00; healing SP 81.99; output SR 74.94
V12	Crit 17.4 (6); AtkPercent 2.7 (1); Spd 2.7 (1); DefFlat 4.5 (1)	6, CritDmg, +15	output	7.6000	89.03	yes	SP	96.7	21.3	—	output SP 89.03; healing SP 81.99
V13	CritDmg 23.2 (6); AtkPercent 2.7 (1); Spd 2.7 (1); HpFlat 110.0 (1)	6, Crit, +15	output	7.6000	89.03	yes	SP	96.7	21.3	—	output SP 89.03; healing SP 81.99
V14	Crit 8.1 (3); Spd 5.4 (2);	4, AtkPercent, +15	output	5.4000	71.80	no	SR	45.0	52.4	—	output SR 71.80

Id	Soul	Slot, main, level	Best fit	U	Score	Specialized	Tier	Depth	Breadth	Growth	Every candidate
	AtkPercent 2.7 (1); HpFlat 300.0 (3)										
V15	Crit 8.4 (3); Spd 5.4 (2); AtkPercent 2.7 (1); HpFlat 300.0 (3)	4, AtkPercent, +15	output	5.5000	72.59	no	SR	46.7	51.5	—	output SR 72.59
V16	AtkFlat 72.0 (3); HpFlat 205.0 (2); DefFlat 13.5 (3); CritDmg 3.6 (1)	4, EffectHit, +15	hit	0.0000	0.00	no	N	0.0	0.0	—	hit N 0.00
V17	Spd 5.7 (2); Crit 5.6 (2); AtkPercent 2.8 (1)	2, Spd, +9	output	4.7000	66.32	no	SR	31.7	59.1	73.58	output SR 66.32; healing R 59.01; hit N 36.03
V18	Spd 11.9 (4); HpPercent 2.7 (1); DefPercent 2.7 (1); AtkFlat 27.0 (1)	2, Spd, +15	hit	5.7667	74.59	no	SR	66.1	30.5	—	hit SR 74.59; healing SR 67.63; output R 60.58
V19	Spd 12.0 (5); HpPercent 2.7 (1); DefPercent 2.7 (1); AtkFlat 27.0 (1)	2, Spd, +15	speed	4.0000	81.24	no	SSR	66.7	n/a	—	speed SSR 81.24; hit SR 74.85; healing SR 67.89; output R 60.84
V20	Spd 12.1 (5); HpPercent 2.7 (1); DefPercent 2.7 (1); AtkFlat 27.0 (1)	2, Spd, +15	speed	4.0333	81.55	no	SSR	67.2	n/a	—	speed SSR 81.55; hit SR 75.12; healing SR 68.15; output R 61.10
V21	Spd 12.0 (4); HpPercent 2.7 (1); DefPercent 2.7 (1); AtkFlat 27.0 (1)	2, Spd, +15	speed	4.0000	81.24	no	SSR	66.7	n/a	—	speed SSR 81.24; hit SR 74.85; healing SR 67.89; output R 60.84
V22	Spd 14.4 (5); HpPercent 2.7 (1); DefPercent 2.7 (1); AtkFlat 27.0 (1)	2, Spd, +15	speed	4.8000	88.74	no	SSR	80.0	n/a	—	speed SSR 88.74; hit SSR 81.14; healing SR 74.15; output SR 67.10
V23	Spd 15.0 (6); HpPercent 2.7 (1); DefPercent 2.7 (1); AtkFlat 27.0 (1)	2, Spd, +15	speed	5.0000	90.62	no	SSR	83.3	n/a	—	speed SSR 90.62; hit SSR 82.71; healing SR 75.72; output SR 68.67
V24	Spd 16.2 (6); HpFlat 100.0 (1); DefFlat 4.5 (1); AtkFlat 25.0 (1)	4, EffectRes, +15	speed	5.4000	94.37	yes	UR	90.0	n/a	—	speed UR 94.37
V25	Spd 18.0 (6); HpPercent 2.7 (1); DefPercent 2.7 (1); AtkFlat 27.0 (1)	2, Spd, +15	speed	6.0000	100.00	yes	UR	100.0	n/a	—	speed UR 100.00; hit SP 90.57; healing SP 83.55; output SSR 76.50
V26	Spd 8.1 (3); Crit 5.4 (2); HpPercent 2.7 (1); DefFlat 5.0 (1)	4, EffectRes, +15	—	—	—	—	unrated	—	—	—	none
V27	HpPercent 8.4 (3); CritDmg 7.6 (2); Spd 5.7 (2); DefFlat 5.0 (1)	6, Crit, +15	healing	6.6000	81.20	no	SSR	46.7	63.1	—	healing SSR 81.20; output R 59.27
V28	HpPercent 11.1 (4); Crit 5.7 (2); Spd 2.7 (1); AtkFlat 27.0 (1)	4, HpPercent, +15	healing	6.5000	80.42	no	SSR	61.7	44.4	—	healing SSR 80.42; hit SR 65.43

Id	Soul	Slot, main, level	Best fit	U	Score	Specialized	Tier	Depth	Breadth	Growth	Every candidate
V29	EffectHit 13.6 (4); Spd 5.4 (2); HpPercent 2.7 (1); DefFlat 5.0 (1)	4, EffectHit, +15	hit	6.1000	77.21	no	SSR	56.7	46.1	—	hit SSR 77.21
V30	Spd 13.5 (5); HpPercent 2.5 (1)	2, Spd, +12	speed	4.5000	85.93	no	SSR	75.0	n/a	86.72	speed SSR 85.93; healing SR 71.28; hit SR 71.19; output SR 64.75

Id	What it shows
V01	obviously poor: flat lines and one EffectRes increment, useful to no v2 archetype (resist, 17.07, in v1.1)
V02	ordinary useful: three useful increments
V03	strong balanced SSR, no specialized line
V04	one perfect line, nothing useful beside it: SSR, not SP (SR in v1)
V05	valid SP
V06	near SP: five roll units exactly is not specialized
V07	specialized, below the SP floor: SSR, not SP (SR in v1)
V08	valid UR: the paper's theoretical output soul
V09	valid UR without a specialized line
V10	near UR: eight useful increments at maximum, one wasted
V11	Speed-focused: UR under speed, SP under hit (hit SP in v1.1)
V12	Crit-focused
V13	CritDmg-focused: the counterpart of V12
V14	dominated: V15 improves its Crit
V15	dominates V14
V16	rare main, useless lines: a rare main earns nothing
V17	growth: +9, two rolls left
V18	speed gate, just below: 3.97 roll units, speed is not a candidate
V19	speed gate, exactly four roll units: speed is a candidate
V20	speed gate, just above
V21	four roll units from four maximal increments: eligible by value, like V19
V22	strong Speed: 4.8 roll units
V23	speed's UR boundary: exactly five roll units is SSR
V24	extreme Speed on an EffectRes main: UR under speed, its only candidate
V25	the theoretical maximum Speed line: 100 under speed
V26	unrated: an EffectRes main with Speed below the gate has no candidate archetype
V27	healing: SSR under healing, R under output on the same soul
V28	healing on an HpPercent main: SSR under healing, SR under hit
V29	hit: EffectHit and Speed, equal weights
V30	growth under speed: +12, eligible now, one roll left

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